... Lecture 7 ...
Genome Wide Markers & EBVs

Recommended Reading:
VanRaden PM. 2008. Efficient methods to compute genomic predictions. *Journal of Dairy Science*, *91*(11), 4414-4423. (see Learn)

Some useful wider reading for the statistically minded "The Elements of Statistical Learning" by Hastie, Tibshirani & Friedman – if I did another statistics qualification I would want it based on this text book! With increasing complexity of data, the range of models open to a geneticist is becoming very wide. How to develop and test such models has developed considerably and this book is very readable, full of examples and written by experts!

In what follows we are assuming markers are bi-allelic SNP throughout. *We shall also omit explicit reference to the nuisance factors, fixed or random. These can be included in models as required, model descriptions and fitting for such terms does not change!*

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Part 1: a different approach

Revision

- Examined polygenic 'infinitesimal' models
  - » modelled by **A** via pedigree
- Single identified locus
  - » always include a polygenic effect from pedigree
    - » if no 'polygenic' the bias in QTL effects are the same as the stratification problems that underlie GWAS
  - » locus as fixed effect to estimate an average effect
  - » locus as random effects to estimate a variance
  - » calculate total EBV for genetic risk or selection

In the previous lectures our attention has been on (i) estimating breeding values without markers, or (ii) estimating the effect of a single locus (or maybe a handful) that is *specified* either by state i.e. where genotypes are known, or by position, which contribute only a part of the total genetic variance. This information from the single locus was combined with the remaining pedigree information to provide a combined estimate of breeding value (EBV). Even where multiple markers were used, the purpose was to help identify IBD for the specified position of the single locus.

Some Limitations

- Difficult to extend to > handful of loci
- What if I don't have pedigree?
  - » e.g. typical medical or wildlife data
- What if there are huge amounts of data?
  - » SNP chips deliver 100,000's markers
  - » sequence delivers 100 millions variants
    - » sequence has all variants in the data

Much of the technology described above relies on the availability of an informative pedigree. This is not easily attainable in medical genetics, or in some wild species although in some populations it can be reconstructed using markers. However the technology does provide basic concepts for further development such as genetics as random effects, estimation of the variance, and predicting EBVs. It can even encompass the idea of a QTL and estimation of its effects. However limitations come when many loci need to be incorporated.

Opportunities

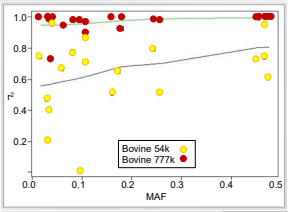
- Can markers capture all variance?
- With large numbers of markers genome wide
  - » maybe every variant is in population-wide linkage disequilibrium with a number of markers
  - » maybe genotype at a trait locus can be predicted from one or more marker loci
- Assume this is the case for now!
  - » i.e. the markers can capture all the genetic variance

As markers become denser, and as we contemplate sequence information as routine, one might consider that the collection of markers would be capable of capturing all the genetic variance. With sequence the variant is contained in the data. with dense markers population linkage disequilibrium should become strong enough to predict genotypes at the variants underlying the genetic variance. This might offer a route to avoid the use of pedigree. With sufficient information for training i.e. estimating SNP effects, one might predict a breeding value, genetic value, without a phenotype at any stage in life from conception.

Predicting Genotypes

Marker SNP within 500 kb of target

Target & marker loci coded 0,1,2



This is a feasible proposition. The graph shows the accuracy of predicting a SNP genotype (defined 0, 1 or 2) from neighbouring loci at loci of different MAF simply by multiple regression. The data come from ~ 1000 genotyped cattle. It shows that for a sufficiently high density neighbouring loci are jointly capable of predicting with very high accuracy, but accuracy is notably lower with a less dense chip.

Here for the first part of what follows we will assume that the markers are capable of describing all the genetic variance, and we will return to that issue later.

Opportunities

- Can predict additive effects of a QTL locus from additive effects of marker loci
  - » use multiple markers not a single locus
  - » HD better prediction particularly for low MAF?
- So with enough markers, all variants can be (nearly) explained by regressions on markers
- Can dispense with 'polygenic variance'
  - » will not require pedigree!

With this assumption we can dispense with the need for 'polygenic variance' associated with pedigree. By our assumption we can see our genetic variance in the data, we do not need any help from the pedigree to give it structure to help fill in the missing bits.

Marker Model (1)

- m markers, $j = 1, \ldots m$
- n records, y_i, $i = 1, \ldots n$
  - » assume for now all records are distinct individuals
  - » assume all individuals with records have genotypes for all markers
- Allele frequency in records is p_j for marker j
- Individual i has x_{ij} copies of alternative allele at j
  - » *i.e.* x_{ij} = 0, 1 or 2, form an n x m design matrix, **X**

So let us consider a marker-only model with m markers and n individuals. For now assume all the records are on distinct individuals, and all that have records have genotypes, and no genotypes are missing.

Let the frequency of a randomly chosen reference allele at locus j be p_j in the data. Define a n x m matrix **X** which has elements x_{ij} which maps the number of alternative alleles at locus j carried by individual i, so x_{ij} = 0, 1 or 2. The mean of x_{ij} over all individuals i is $2p_j$.

Marker Model (2)

- Marker j has true effect of allelic substitution b_j in population
  - » form an m x 1 vector **b**
- Breeding value can be explained by markers
  $\mathbf{g} = \mathbf{Xb}$ (i.e. $g_i = \sum_{loci} x_{ij} b_j$)
  - » where **g** is a n x 1 vector of breeding values
- If have estimates of marker effects $\hat{\mathbf{b}}$
  $\hat{\mathbf{g}} = \mathbf{X}\hat{\mathbf{b}}$ (i.e. $\hat{g}_i = \sum_{loci} x_{ij} \hat{b}_j$) are EBV's

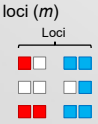
Assume locus j has true effect b_j. Think about the b_j as a regression of the merits on the 0/1/2's as illustrated in Falconer, then the b_j's are effects of allelic substitutions (see later). The true breeding value (TBV) can then be obtained by summing, over all loci, the true effects times the number of alleles, i.e. $g_i = \sum_{loci} x_{ij} b_j$ or $\mathbf{g} = \mathbf{Xb}$. Likewise the EBV uses the estimated effects rather than the true effects. Note the matrix notation, try and use it.

Small Example

- **X** for 3 individuals (n) with 2 loci (m)

Loci

Individuals $\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \end{pmatrix}$



☐ is reference allele for a locus
Alternative alleles:
Locus 1, allele effect ■ = (☐ +2)
Locus 2, allele effect ■ = (☐ +1)

Breeding values		
①	g_1	+4
②	g_2	+1
③	g_3	+6

A small example to demonstrate the TBV is obtained by the matrix formula.

g = X b

Breeding values		
①	g_1	+4
②	g_2	+1
③	g_3	+6

$$\begin{matrix} g_1 \\ g_2 \\ g_3 \end{matrix} = \begin{matrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \end{matrix} \begin{matrix} b_{■} \\ b_{■} \end{matrix} \Rightarrow \begin{matrix} 4 \\ 1 \\ 6 \end{matrix} = \begin{matrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \end{matrix} \begin{matrix} 2 \\ 1 \end{matrix}$$

g = **X** **b**

Multiple Regression

- Model $\mathbf{y} = \mathbf{1}\mu + \mathbf{Xb} + \mathbf{e}$
  - » **y** is (n x 1) vector of observations
  - » **1**μ is a mean vector, with **1** (n x 1) vector of 1's
  - » **b** is (m x 1) vector of fixed marker effects
  - » **X** is (n x m) design matrix, mapping markers to individuals
  - » **e** is (n x 1) vector of residual effects
    - » **e** is assumed to be MVN(0, σ_E^2**I**)
- Above ignores nuisance factors
  - » know models can handle them

Notation!
μ is mean,
m is # markers.

Here μ is replacing m as the mean in the notation, because m is the number of markers!

So one might think that carrying out a multiple regression on the markers will then capture the variance. However, unfortunately this becomes impossible since the number of markers available or required to capture the variance is far greater than the number of data points available for estimating effects. This means that the key matrix $\mathbf{X}^T\mathbf{X}$ is not invertible, as it needs to be for obtaining solutions in least squares models. This is before *any* consideration of whether or not multiple regression (a least squares model) would provide the best predictor.

Multiple Regression

- Model $\mathbf{y} = \mathbf{1}\mu + \mathbf{X}\mathbf{b} + \mathbf{e}$
  - » **y** is (n x 1) vector of observations
  - » **1**μ is the mean vector, with **1** (n x 1) vector of 1's
  - » **b** is (m x 1) vector of regression coefficients mapping markers to
  - » **e** is (n x 1) vector of residual effects
    - » **e** is assumed to be MVN(0, σ_E^2**I**)
- So $\hat{\mathbf{b}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$ and $\hat{\mathbf{g}} = \mathbf{X}\hat{\mathbf{b}}$

BUT m > n … more predictors than records
No inverse! No solution! Over-fitting!

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In the world of omics, this a typical problem where there are more predictors than data points, as our ability to measure outstrips our ability to make use of data efficiently! Here the number of markers m is > and often > > the number of data points n.

Predictors >> Data

- An increasingly common problem
  - » e.g. 100,000s markers, 1000's phenotypes
  - » 'omics data generates yet more 100,000s of expression data to fit the same 'few' phenotypes!
- Options
  - » reduce dimensionality
  - » subset selection
  - » shrinkage

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We have already come across this problem in the course: e.g. when one predicts a polygenic breeding value for every individual in a pedigree with other fixed effects, or an EBV for every individual for polygenic variance and QTL variance, the number of terms being estimated has exceeded the number of data points.

There are methods to overcome this problem. For example we have used models assuming random effects which use shrinkage, i.e. regression back to the mean, depending on information available. There are other methods which can involve reducing dimensionality e.g. principle components or partial least squares, or a related idea of subset selection. We will only briefly touch on the latter in QGM.

Ridge Regression

- Established method for overcoming $m > n$ in regression problems
  - » assume the m regression coefficients on the marker loci are a random effect (i.e. the b's are random)
  - » 'shrinkage' method, shrinks coefficients towards 0
  - » shrinkage at each marker will depend on info
  - » can formulated as a penalty on the size of b_j's i.e. $\mathbf{b}^T\mathbf{b} = \sum_{loci} b_j^2$
  - » we will stick to mixed model formulation

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Ridge regression is an established method for when predictors are large in comparison to data points. *The key idea in ridge regression is that the set of regression coefficients on the marker loci are treated as a random effect.* Therefore it is a shrinkage method that says if there is not much information don't allow it to have a big impact so make the regression coefficient small.

In a 'frequentist' setting it can be developed as an optimisation of a fit which reduces the residual sum of squares to the maximum possible degree whilst suffering a penalty on the sum of squares of the regression coefficients. Therefore a large coefficient is a big penalty that must prove its worth in explaining a lot of variance.

Centre the X

- In simple regression it is easiest to make regression slope independent of the intercept
  - » computation ease, statistical clarity
  - » does not change model fits when fixed effect
  - » 'centre' covariates as deviations from their mean
  - » e.g. $\mathbf{y} = \mathbf{1}a + \mathbf{x}b + \mathbf{e}$ becomes $\mathbf{y} = \mathbf{1}\mu + (\mathbf{x} - \mathbf{1}\bar{x})b + \mathbf{e}$
- This is typically done in ridge regression
- Here, **X** is (n x m) design matrix mapping the m markers to the n individuals

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To develop the topic it is useful to simplify the statistics and computation by centring each covariate by subtracting its mean; this does not change the fit (i.e. fitted values will be the same).

In our problem the covariates are the marker genotypes, which in the past lectures have been assumed to be coded 0 for the homozygote of a randomly chosen reference allele, 1 for the heterozygote, and 2 for the alternative homozygote.

Centre the X

- 'Centre' **X** ➡ $\mathbf{W}_1$

Loci

Individuals $\begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 2 & 2 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & 1/3 \\ -1 & -2/3 \\ 1 & 1/3 \end{pmatrix}$

Frequency, p_j	½	5/6
Mean of covariate, $2p_j$	1	5/3

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We started with the design matrix mapping individual records (rows) to markers (columns) as **X**, now it will be denoted $\mathbf{W}_1$ as it has been centred. The mean of the alternative allele counts for locus i is $2p_i$ where p_i is the frequency of the alternative allele (i.e. the allele that has been counted!).

SNP BLUP (1)

- Model $\mathbf{y} = \mathbf{1}\mu + \mathbf{W}_1\mathbf{b} + \mathbf{e}$
  - » **y** is (n x 1) vector of observations
  - » **1**μ is the mean vector, with **1** (n x 1) vector of 1's
  - » **b** is (m x 1) vector of random marker effects
    - » **b** assumed to be MVN(0,σ_b^2**I**)
    - » random regression coefficients with mean 0
  - » $\mathbf{W}_1$ is (n x m) design matrix, mapping centred markers (columns) to individuals (rows)
  - » **e** is (n x 1) vector of residual effects
    - » **e** assumed to be MVN(0,σ_E^2**I**)

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Therefore ridge regression is a mixed model which places a prior distribution on the regression coefficients with a particular variance that may need estimation. It is an example of a 'random regression' model, by seeing these coefficients as a random effect we can estimate them even if $m \gg n$.

When applied to estimating breeding values using a large number of SNP loci, the model is often referred to as 'SNP BLUP'. The key term is that **b**, the vector of m regression coefficients, is assumed to be MVN(0, σ_b^2**I**); so the true coefficients of markers i and j are assumed to be independently sampled from a Normal distribution mean 0, variance σ_b^2.

SNP BLUP (1)

- Solutions are $\hat{\mathbf{b}} = (\mathbf{W}_1^T\mathbf{W}_1 + \gamma^{-1}\mathbf{I})^{-1}\mathbf{W}_1^T\mathbf{y}$
  - » can be inverted (as add a constant to diagonal)
  - » where $\gamma = \sigma_b^2/\sigma_E^2$
  - » σ_b^2 estimated by REML, or assumed known
- $\hat{\mathbf{g}} = \mathbf{W}_1\hat{\mathbf{b}}$ are EBV's
  - » can give EBV for any individual with genotypes
    - » even if not used to estimate parameters

So solutions follow standard mixed model forms. The solution can be obtained because the ridge regression assumption is equivalent to adding a constant (σ_b^2 / σ_E^2) to the leading diagonal of the $\mathbf{W}_1^T\mathbf{W}_1$ matrix which makes this matrix, full rank and invertible.

The EBVs come from using the estimated SNP effects for any individual with genotypes, including individuals that have genotypes but have not been used to estimate the SNP effects.

SNP BLUP (1)

- Genetic interpretation of regressions on 0,1,2
  - » effects of allelic substitution (α) i.e. $b_j = \alpha_j$ if
    - » loci assumed strictly additive i.e. $d_j = 0$;

    or
    - » population used for estimation is a sample of target population in which estimates will be used
      - » so allele frequency at each locus is similar
- No pedigree, or map or map order is needed

The variance of coefficients, σ_b^2, can be viewed as a variance of allelic substitution effects, particularly if the traits are assumed completely additive with no dominance, or if the population analysed is a representative sample (which excludes a case control population!) of the target population for use. This interpretation of placing a variance on allelic effects can be considered 'physiological', concerned with the impact of mutations on gene expressions. Therefore the variance makes no reference to MAF at loci.

Note no pedigree, or map, or map order is required for the model.

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Part 2: from SNPs to Breeding Values

Exploring Breeding Values

- What are Estimated Breeding Values (EBV)?
  $\hat{\mathbf{g}} = \mathbf{W}_1\hat{\mathbf{b}}$
- What are True Breeding Values (TBV)?
  $\mathbf{g} = \mathbf{W}_1\mathbf{b}$
  - » assumed markers can precisely predict the TBV
- $E[\mathbf{g}] = E[\mathbf{W}_1\mathbf{b}] = \mathbf{W}_1E[\mathbf{b}] = 0$
  - » as regression coefficients were assumed to be MVN(0,$\sigma_b^2\mathbf{I}$) i.e. have mean 0
  - » i.e. mean allelic substitution effect is 0

Having fitted this model the EBV would be obtained as $\mathbf{W}_1\hat{\mathbf{b}}$ i.e. the sum of the estimates for the alleles that it carries. This follows from the model of the TBV i.e. $\mathbf{g} = \mathbf{W}_1\mathbf{b}$. Now note that in the model E[**b**] = 0 and so E[**g**] = 0. (This is just the breeding value term, in the model of data we will have the mean and the other fixed effects.)

Covariance of Breeding Values

- The (co)variance matrix of the TBV

$$E[\mathbf{g}\mathbf{g}^T] = E[\mathbf{W}_1\mathbf{b}(\mathbf{W}_1\mathbf{b})^T] = E[\mathbf{W}_1\mathbf{b}\mathbf{b}^T\mathbf{W}_1^T]$$
$$= \mathbf{W}_1E[\mathbf{b}\mathbf{b}^T]\mathbf{W}_1^T = \sigma_b^2\mathbf{W}_1\mathbf{I}\mathbf{W}_1^T$$
$$= \sigma_b^2\mathbf{W}_1\mathbf{W}_1^T$$

- $\mathbf{W}_1\mathbf{W}_1^T$ is an (n x n) matrix of the scaled (co)variances of all TBV in our population
  - » scaled by σ_b^2
  - » its definition is very similar to definition of **A**!

The (co)variance of the true breeding values (TBV) for individuals can be expressed as E[$\mathbf{g}\mathbf{g}^T$].

$$E[\mathbf{g}\mathbf{g}^T] = E[\mathbf{W}_1\mathbf{b}(\mathbf{W}_1\mathbf{b})^T] = E[\mathbf{W}_1\mathbf{b}\mathbf{b}^T\mathbf{W}_1^T]$$
$$= \mathbf{W}_1E[\mathbf{b}\mathbf{b}^T]\mathbf{W}_1^T = \sigma_b^2\mathbf{W}_1\mathbf{I}\mathbf{W}_1^T = \sigma_b^2\mathbf{W}_1\mathbf{W}_1^T$$

This expresses the (co)variance of breeding values in terms of the genomic data. It looks a little like **A**: $E[\mathbf{g}\mathbf{g}^T] = \sigma_A^2\mathbf{A}$. Remember (co)variances are relationships, so $\sigma_b^2\mathbf{W}_1\mathbf{W}_1^T$ is a kind of relationship matrix but here the scaling variance σ_b^2 is the variance of the individual marker allelic substitution effects.

Covariance of Breeding Values

- However in **A** the scaling variance is a measure of additive genetic variance
  - » in the base population
- In $\mathbf{W}_1\mathbf{W}_1^T$, it is a variance of individual marker allelic substitution effects σ_b^2
  - » if change m would expect σ_b^2 to change
  - » ↓ m, fewer markers trying to explain the QTL
  - » ↑ m, spread genetic variance over more loci
  - » remember the loci are anonymous markers

However the scaling of this relationship is about the variance of allelic substitution effects per locus. This is not directly of as much interest as having an idea of the genetic variance in a trait. Also to some extent this variance will depend on the number of markers we have as what is being done is to share the entire variance, so this value will depend on how many markers we have to explain the variance ... its value will be related to 1/m in some way. Remember these are anonymous markers.

Covariance of Breeding Values

- Suppose scale $\mathbf{W_1}$ by a constant c
  - » i.e. $\mathbf{W}^* = \mathbf{W}_1/c$ and $\mathbf{W}_1 = c\mathbf{W}^*$
  - » then $E[\mathbf{gg}^T] = \sigma_b^2(c\mathbf{W}^*)(c\mathbf{W}^{*T}) = c^2\sigma_b^2\mathbf{W}^*\mathbf{W}^{*T}$
- Choose $c = \sqrt{\sum_{loci} 2p_j(1-p_j)}$
- Then $E[\mathbf{gg}^T] = (\sum_{loci} 2p_j(1-p_j)\sigma_b^2)(\mathbf{W}_1\mathbf{W}_1^T / c^2)$

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So one way of obtaining a more useful variance is to scale the $\mathbf{W}_1$-matrix. If we scale $\mathbf{W}_1$ by *dividing* all its elements it by a value c, then the variance estimated is scaled by *multiplying* it by c^2. So choose $c = \sqrt{\sum_{loci} 2p_j(1-p_j)}$ and calculate $w^*_{ij} = w_{1,ij} / c$. i.e. $\mathbf{W}^* = \mathbf{W}_1 / c$.

So $E[\mathbf{gg}^T] = c^2\sigma_b^2(\mathbf{W}_1\mathbf{W}_1^T / c^2) = (\sum_{loci} 2p_j(1-p_j)\sigma_b^2)(\mathbf{W}_1\mathbf{W}_1^T / c^2)$.

Covariance of Breeding Values

- Let $\sigma_g^2 = \sum_{loci} 2p_j(1-p_j)\sigma_b^2$
  - » looks like a genetic variance, $2pq\alpha^2$
  - » summed over loci, replacing α^2 by $E[\alpha^2]$, i.e. σ_b^2
  - » it is the genic variance
    - » it is additive genetic variance of the individuals used to construct $\mathbf{W}$ if all loci are in HWE
- So $\mathbf{g}$ is distributed $E[\mathbf{g}] = 0$ with (co)variances

$$E[\mathbf{gg}^T] = (\sum_{loci} 2p_j(1-p_j)\sigma_b^2)(\mathbf{WW}^T / c^2)$$
$$= \sigma_g^2(\mathbf{WW}^T / c^2) = \sigma_g^2\mathbf{G}_1$$

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The variance of a single locus is $2p_j(1-p_j)\alpha_j^2$ assuming Hardy-Weinberg equilibrium (HWE), and $E[\alpha^2] = \sigma_b^2$; so summed over many loci, $\sigma_g^2 = c^2\sigma_b^2$ looks like a genetic variance. However as variance is simply summed over all loci ignoring LD, this represents a genetic variance for all loci being in HWE, and in linkage equilibrium (LE). It is called the genic variance. (It will be shown later that this σ_g^2 is also the estimate of the genetic variance in the population used to construct $\mathbf{W}_1$ if in HWE). This $\mathbf{G}_1 = \mathbf{W}^*\mathbf{W}^{*T} = \mathbf{W}_1\mathbf{W}_1^T / c^2$ is called a genomic relationship matrix.

Note this $\mathbf{G}_1$ is defined entirely by the markers carried by the individuals in the data and has no pedigree information. It is a genomic relationship matrix. It relies on IBS (identity by state) not IBD (identity by descent). $\mathbf{G}_1$ is a n x n matrix where n is the number of individuals used to construct $\mathbf{W}_1$ and σ_g^2 refers to this population – we will return to this.

GBLUP Van Raden Model 1

- Model $\mathbf{y} = \mathbf{1}\mu + \ldots + \mathbf{Zg} + \mathbf{e}$
  - » $\mathbf{y}$ is (d x 1) vector of observations
  - » $\mathbf{1}\mu$ is the mean vector, with $\mathbf{1}$ (d x 1) vector of 1's
  - » … [nuisance effects]
  - » $\mathbf{g}$ is (n x 1) vector of breeding values
    - » $\mathbf{g}$ assumed to be random and distributed MVN(0,$\sigma_g^2\mathbf{G}_1$)
  - » $\mathbf{Z}$ is (d x n) design matrix for $\mathbf{g}$
  - » $\mathbf{e}$ is (d x 1) vector of residual effects
    - » $\mathbf{e}$ assumed to be MVN(0,$\sigma_E^2\mathbf{I}$)

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With this new relationship matrix we can construct a new, but equivalent model. We have derived, starting from ridge regression, a genomic BLUP model (GBLUP), which fits a genomic relationship matrix to the observed data.

It is known as Van Raden Model (1); it is Model (1) because of the construction and scaling of the $\mathbf{G}$-matrix.

Note there are d individuals with records, with n individuals used to form the genomic relationship matrix, and (just like the pedigree models) $d \leq n$. All individuals with records must be in $\mathbf{G}_1$.

A summary of how to calculate $\mathbf{G}_1$.

Van Raden Model 1

- Derived GBLUP (with Van Raden Model 1)
- Uses a relationship matrix $\mathbf{G}_1$
  - » calculated from markers alone $\mathbf{G}_1 = \mathbf{W}_1\mathbf{W}_1^T / c^2$
    1. form $\mathbf{W}_1$ as centred allele counts
    2. form $\mathbf{W}_1\mathbf{W}_1^T$
    3. scale by $c^2 = \sum_{loci} 2p_j(1-p_j)$

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Van Raden Model 1

- Has variance on allelic substitution effects
  - » variance σ_g^2 is a genic variance
  - » additive genetic for individuals used to construct $\mathbf{W}_1$, if loci in HWE
- All loci are scaled by same value c
- Scaling by c does not alter the EBV obtained
  - » the scaling of $\mathbf{W}_1$ only changes the value of 'the genetic variance', σ_g^2
  - » does not change (co)variances of breeding values as c values in σ_g^2 and $\mathbf{W}_1\mathbf{W}_1^T/c^2$ cancel

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An important point to note here. All values of the centred design matrix $\mathbf{W}_1$ are scaled by the same value. Changing this value, changes the variance component we obtain as a parameter, however it does not change the estimate of the genetic (co)variances, genomic EBV, and hence the accuracy of the evaluation.

Van Raden Model 1

- It is the same model as SNP BLUP (1)
  - » only expressed in a different form
- All individuals used for deriving $\mathbf{G}_1$ have EBV
  - » irrespective of whether they have records

Van Raden Model 1 is an equivalent model to SNP BLUP 1.

Note that all individuals in $\mathbf{G}_1$ get an EBV even if they have no record.

Van Raden Model 1

- Variance explained by a locus is

$$2p_j(1-p_j)\alpha_j^2$$

- What is expected variance explained for a locus with allele frequency p ?

$$E[2p_j(1-p_j)\alpha_j^2] = 2p_j(1-p_j)E[\alpha_j^2] = 2p_j(1-p_j)\sigma_b^2$$

- It varies with p …
  - » intermediate frequencies highest per locus
  - » MAF close to 0 contribute very little

The expected variance explained by locus j depends on p_j is $2p_j(1-p_j)\sigma_b^2$; so intermediate frequencies are expected to be associated with the highest variance per locus and those with MAF close to 0 associated with very little variance.

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Part 3: another **G**!

Ridge Regression

- Treated allelic substitutions as random effect
  - » distribution assumptions were defined independent of MAF
- Effects are regressed back to zero …

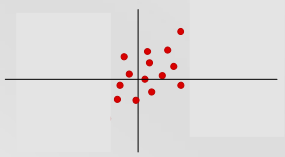
In ridge regression, regression coefficients are treated as random and so the coefficients are shrunk towards 0, with the amount of shrinkage depending on the amount of information.

Ridge Regression

- Regression to zero ('shrinkage') greatest
  - » if σ_b^2/σ_E^2 is small
    - » attribute of the problem, same for all loci
  - » if uncertainty over locus effect is greatest
    - » sampling error for locus j is $\sim 1/\sum_{individuals\ i} w_{ij}^2$
    - » $\sum_i w_{ij} = 0$ as centred, so $\sum_i w_{ij}^2 = n\,var(w_{ij})$ for locus j
    - » so uncertainty greatest when $var(w_{ij})$ is small

One factor leading to large shrinkage is when σ_b^2 is small in relation to σ_E^2, but this will be the same for all marker loci in the analysis. The other factor is how much information is available; those loci with least information will be shrunk most.

How confident about the slope?

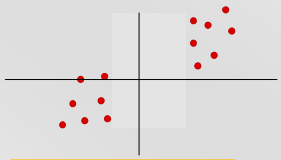


Narrow spread of x's – small variance

Recall the sampling variance of a regression coefficient estimated by least squares, $\sigma_e^2/\sum(x-\bar{x})^2$, where the denominator is related to the variance of the covariate. It is a measure of how confident we are in the estimate of the coefficient.

To demonstrate this, look in this and next slide consider which one gives you the best idea if the slope.

How confident about the slope?



Wide spread of x's – large variance

It is the spread of the x-values – the variance – that gives more confidence.

Look back at the heuristic for random effects uploaded for Lecture 5. Therefore those loci with least variance will be shrunk most. The variance at a locus is ~ 2p(1-p) assuming Hardy Weinberg, so the loci with the lowest allele frequencies that will tend to be shrunk most.

Ridge Regression

- Genetically
  - » hypothesis of rare alleles of large effect
  - » so genetic variance resides in low MAF
- Statistically
  - » it is common to standardise design matrix
  - » divide regression variables by s.d.
  - » centred and normalised

It can be argued the largest effects tend to be associated with low MAF and of course the distribution of p_j is U-shaped. So more variance arises from loci with low MAF. Beware – the genotypes we are considering are just anonymous markers not causal mutations so this argument does not apply – what is being done is trying to predict each QTL with many SNPs and the best regressions come from more informative loci.

Statistically it is common when carrying out ridge regression normalise the covariates by dividing through by the s.d. of the covariate. This is not done in Van Raden Model (1). For example it is sensible if the problem involves developing a predictor from a number of covariates all measured in different units, and normalising overcomes these problems. In the SNP case, this justification is less relevant, as all covariates are SNPs, so is it reasonable to normalise the SNP by MAF?

Centre the X

- 'Centre' **X** ➡ $\mathbf{W}_1$

$$\text{Individuals}\left[\overbrace{\begin{pmatrix}1 & 2\\ 0 & 1\\ 2 & 2\end{pmatrix}}^{\text{Loci}}\right] \Rightarrow \overbrace{\begin{pmatrix}0 & 1/3\\ -1 & -2/3\\ 1 & 1/3\end{pmatrix}}^{\text{Loci}}$$

Frequency, p_j	½	5/6
Mean of covariate, $2p_j$	1	5/3

The extra step: *dividing* through each column of $\mathbf{W}_1$ by its s.d. assuming HWE, i.e. $\sqrt{2p_j(1-p_j)}$. *So a HWE assumption is appearing here*!

Note dividing through by the s.d. is called 'normalising'.

Denote this normalised matrix by $\mathbf{W}_2$. However, normalising $\mathbf{W}_1$ affects the idea that the breeding value is the simple sum of the alleles an individual carries based on the allelic substitution effect! Therefore in order to retain this fundamental idea, if $\mathbf{W}_1$ is scaled then a balancing scaling must be carried out to the regression coefficients.

Normalise $\mathbf{W}_1$

- Normalising $\mathbf{W}_1$ ➡ $\mathbf{W}_2$

$$\text{Individuals}\left[\overbrace{\begin{pmatrix}0 & 1/3\\ -1 & -2/3\\ 1 & 1/3\end{pmatrix}}^{\text{Loci}}\right] \Rightarrow \begin{pmatrix}0.00 & 0.63\\ -1.41 & -1.26\\ 1.41 & 0.63\end{pmatrix}$$

Frequency (p_j)	½	5/6
s.d. ~ $[2p_j(1-p_j)]^{1/2}$	0.71	0.53

Assumes HWE

So the b_j's are transformed to β_j's by *multiplying* $\sqrt{2p_j(1-p_j)}$ i.e. they no longer represent allele substitution effects, they are scaled substitution effects with the scaling depending on p_j. The variance of the β_j's is $E[\beta_j^2]$, is then the expectation of $2p_j(1-p_j)\alpha_j^2$; we recognised this as an expression of genetic variance for locus *j*. So $E[\beta_j^2]$ is now a contribution to genetic variance.

Normalise $\mathbf{W}_1$

- If the regression variables are scaled the slopes b_j will be scaled in opposite way!
  - » $\beta_j = b_j\sqrt{2p_j(1-p_j)}$
  - » so $E[\beta_j^2] = 2p_j(1-p_j)E[b_j^2] = 2p_j(1-p_j)\sigma_b^2$
  - » if b_j's are allelic effects, β_j^2 are contributions to genetic variance
- So if assume $\beta_j \sim MVN(0,\sigma_v^2\mathbf{I})$
  - » implication is genetic variance is <u>expected</u> to be shared equally amongst all marker loci
  - » distribution now depends on MAF!

The model equation follows in a similar fashion to before, only the design matrix and distributional assumptions have changed. Any additional fixed and random effects are not shown here for simplicity.

Genetically the model has changed: (i) each locus is *expected* to contribute the same variance irrespective of MAF (say σ_v^2), (ii) the total genic variance, i.e. assuming LE and HWE, is then $m\sigma_v^2$, and (iii) the distribution of effects will now depend on MAF.

SNP BLUP (2)

- Model $\mathbf{y} = \mathbf{1}\mu + \mathbf{W}_2\boldsymbol{\beta} + \mathbf{e}$
  - » **y** is (n x 1) vector of observations
  - » $\mathbf{1}\mu$ is the mean vector, with **1** (n x 1) vector of 1's
  - » **β** is (m x 1) vector of random marker effects
    - » **β** assumed to be $MVN(0,\sigma_v^2\mathbf{I})$
    - » coefficients with mean 0 are scaled substitution effects
  - » $\mathbf{W}_2$ is (n x m) design matrix, mapping centred & <u>normalised</u> markers (cols) to individuals (rows)
  - » **e** is (n x 1) vector of residual effects
    - » **e** assumed to be $MVN(0,\sigma_E^2\mathbf{I})$

Exploring Breeding Values

- What are Estimated Breeding Values (EBV)? $\hat{\mathbf{g}} = \mathbf{W}_2\hat{\boldsymbol{\beta}}$
- What are True Breeding Values (TBV)? $\mathbf{g} = \mathbf{W}_2\boldsymbol{\beta}$
  - » assumed markers can precisely predict the TBV
- E[**g**] = E[$\mathbf{W}_2\boldsymbol{\beta}$] = $\mathbf{W}_2$ E[$\boldsymbol{\beta}$] = 0
  - » as regressions were assumed to have mean 0

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As before, the EBV would be obtained as $\mathbf{W}_2\hat{\boldsymbol{\beta}}$, i.e. the sum of the estimates for the alleles that it carries. As before, this follows from the model of the TBV i.e. $\mathbf{W}_2\boldsymbol{\beta}$. Now note that in the model E[$\boldsymbol{\beta}$] = 0.

Covariance of Breeding Values

- The covariance of the TBV

$$E[\mathbf{g}\mathbf{g}^T] = E[\mathbf{W}_2\boldsymbol{\beta}(\mathbf{W}_2\boldsymbol{\beta})^T] = E[\mathbf{W}_2\boldsymbol{\beta}\boldsymbol{\beta}^T\mathbf{W}_2^T]$$

$$= \mathbf{W}_2 E[\boldsymbol{\beta}\boldsymbol{\beta}^T]\mathbf{W}_2^T = \sigma_v^2\mathbf{W}_2\mathbf{I}\mathbf{W}_2^T$$

$$= \sigma_v^2\mathbf{W}_2\mathbf{W}_2^T$$

- $\mathbf{W}_2\mathbf{W}_2^T$ is an (n x n) matrix of the (co)variances of all TBV in our population scaled by σ_v^2

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Therefore the covariance between breeding values is $E[\mathbf{g}\mathbf{g}^T] = \sigma_v^2\mathbf{W}_2\mathbf{W}_2^T$. Again, σ_v^2 is a variance per locus so is dependent on the number of markers.

Covariance of Breeding Values

- Remember σ_v^2 is a per locus variance
- Total genic variance is $m\sigma_v^2$
- Let $\sigma_g^2 = m\sigma_v^2$

$$E[\mathbf{g}\mathbf{g}^T] = \sigma_v^2\mathbf{W}_2\mathbf{W}_2^T = (m\sigma_v^2)(\mathbf{W}_2\mathbf{W}_2^T / m) = \sigma_g^2\mathbf{G}_2$$

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We can re-scale using the number of markers, m, to get $E[\mathbf{g}\mathbf{g}^T] = (m\sigma_v^2)(\mathbf{W}_2\mathbf{W}_2^T / m) = \sigma_g^2\mathbf{G}_2$.

Covariance of Breeding Values

- Covariance between breeding values is $\sigma_g^2\mathbf{G}_2$
  - » where $\mathbf{G}_2 = \mathbf{W}_2\mathbf{W}_2^T/m$
    - » the 0/1/2 variates are centred & normalised marker by marker so each has variance 1 across individuals
  - » $\mathbf{G}_2$ is a (n x n) genomic relationship matrix
    - » derived from IBS of markers alone, <u>not IBD</u>
  - » σ_g^2 is a genic variance
    - » additive genetic variance of individuals used to construct **W** if loci are in HWE (more in Lecture 9)

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Now σ_g^2 is the genic variance and then $\mathbf{G}_2$ is a <u>genomic relationship matrix</u>. As with $\mathbf{G}_1$, $\mathbf{G}_2$ is constructed from IBS not IBD.

GBLUP Van Raden Model 2

- Model $\mathbf{y} = \mathbf{1}\mu + \ldots + \mathbf{Z}\mathbf{g} + \mathbf{e}$
  - » **y** is (d x 1) vector of observations
  - » **1**μ is the mean vector, with **1** (d x 1) vector of 1's
  - » … [nuisance effects]
  - » **g** is (n x 1) vector of breeding values
    - » **g** assumed to be MVN(0,$\sigma_g^2\mathbf{G}_2$)
  - » **Z** is (d x n) design matrix for **g**
  - » **e** is (n x 1) vector of residual effects
    - » **e** assumed to be MVN(0,$\sigma_E^2\mathbf{I}$)

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This gives Van Raden Model 2, using $\mathbf{G}_2$ the Van Raden genomic relationship model 2. $\mathbf{G}_2$ is and n x n matrix where n is the number of individuals used to construct it, and **g** is an n x1 vector.

As with Van Raden 1, not all individuals used to form $\mathbf{G}_2$ need have phenotype records, but all will get an EBV.

GBLUP Van Raden Model 2

- $\mathbf{G}_2 = (\mathbf{W}_2\mathbf{W}_2^T)/m$
  - » m = number of markers
- $\mathbf{G}_2 = \mathbf{W}_2\mathbf{W}_2^T/m = \mathbf{W}_1\mathbf{D}^{-1}\mathbf{W}_1^T/m$
  - » **D** is m x m diagonal matrix with $d_{jj}=2p_j(1-p_j)$

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Van Raden Model 2

- GBLUP with Van Raden Model 2
- Uses a relationship matrix $\mathbf{G}_2$,
  - » calculated from markers alone $\mathbf{G}_2 = (\mathbf{W}_2 \mathbf{W}_2^T)/m$
    1. form $\mathbf{W}_1$ as centred allele counts
    2. form $\mathbf{W}_2$ by normalising each locus by dividing by $c_i = \sqrt{2p_i(1-p_i)}$
    3. form $\mathbf{W}_2 \mathbf{W}_2^T$
    4. scale by number of markers m

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A summary of calculating $\mathbf{G}_2$.

Van Raden Model 2

- Has variance on marker effects in 's.d.' units
  - » each locus expected to contribute same variance irrespective of MAF
  - » the estimated variance σ_g^2 is a genic variance
    - » additive genetic variance for individuals used to construct **W** if loci are in HWE
- Implemented in GCTA & ACTA

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A key point is that $\mathbf{G}_2$ is the default usage for GCTA – a widely used medical analysis package.

... Lecture 7 ...
Genome Wide Markers & EBVs

Part 4: some concluding remarks

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Van Raden Models

- Loci counts scaled by same value in Model 1 and locus-specific values in Model 2
- Models 1 and 2 **are** different
  - » variances, effects and EBVs
  - » even if using sequence
  - » distributional assumptions on effects changed
  - » VR(2) has substitution effects related to 1/MAF
- As yet, no evidence of large differences

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In generally use Van Raden Model 1 especially with anonymous SNP markers, but generally differences in outcomes are slight even though the assumptions are different. There is no consensus on which is preferable. However as the number of data records increase the importance of the shrinkage decreases. Remember the sampling variance of a regression coefficient decreases with n var(x) so as n increases the shrinkage decreases. Ultimately the difference between Van Raden Models 1 and 2 will disappear as the prior information on distribution is overwhelmed by the data.

Marker Effects

- Estimates from Van Raden models are EBV
- Can obtain marker effects as
  - » Van Raden Model 1 $\hat{\mathbf{b}} = \mathbf{W}_1^T \mathbf{G}_1^{-1} \hat{\mathbf{g}} / c^2$
  - » Van Raden Model 2 $\hat{\boldsymbol{\beta}} = \mathbf{W}_2^T \mathbf{G}_2^{-1} \hat{\mathbf{g}} / m$
- They are different!
- More in Tutorial 8

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So far we have concentrated on the estimates of variance from the Van Raden models and EBV. What are the effects of the markers? If we are fitting a marker based ridge regression model then the marker effects come directly – and both Van Raden models have an equivalent SNP BLUP model. The marker effects can be recovered using the formulae shown. The proof that this is reasonable is straightforward, the proof that it must be as shown is more difficult and not covered. In Model (1): $\hat{\mathbf{g}} = \mathbf{W}_1 \hat{\mathbf{b}}$, so if $\hat{\mathbf{b}} = \mathbf{W}_1^T \mathbf{G}_1^{-1} \hat{\mathbf{g}} / c^2$ then $\mathbf{W}_1 \hat{\mathbf{b}} = \mathbf{W}_1 \mathbf{W}_1^T \mathbf{G}_1^{-1} \hat{\mathbf{g}} / c^2 = \mathbf{G}_1 \mathbf{G}_1^{-1} \hat{\mathbf{g}} = \hat{\mathbf{g}}$ as it needs to be. Proof of reasonableness for Model (2) is by analogy.

We do more on marker effects in Tutorial 8.

What is σ_g^2?

- σ_g^2 is an 'additive variance' as it is based on regression on allele counts
- EBV will have validity even if σ_g^2 is hard to interpret!
- Can also calculate genetic variance in any subset

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Throughout we have considered the variance σ_g^2 and interpreted what it means. It is different from pedigree based estimates, where there were assumptions concerned with being unrelated. For the genomic relationships, based on IBS, there are much greater demands made for an unrelated population, but it is an estimate of the genetic variance in the population used to construct **G**, providing all the loci were in Hardy-Weinberg equilibrium. So estimates of heritability from genomic analyses may be different in their meaning.

<u>However</u> what we are trying to do is to estimate breeding values using the SNP data, and we are doing this by ridge regression. Pointing to a base population is unnecessary, and worrying about whether the relationships corresponds to our expectations is also irrelevant, providing we are using effective estimation techniques, which is a

statistical problem not a genetic one! We have ways of judging quality which we will explore later.

Using Markers

- Markers may not capture σ_A^2 for trait
  - » consider trait as two components $g_i = u_i + e_i$
    - » g is the trait breeding value
    - » u is the component of the breeding value that is predictable by markers
    - » e is the component independent of the markers, hopefully small
- If I use sequence there <u>may</u> be a prior with effects to be larger for low MAF
  - » if I use markers this does not hold!

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Suppose the markers do not capture all the genetic variance. What I can do is to consider the breeding value as being the sum of a marker-accessible trait (w, which with a very large d I could predict perfectly from the markers) and a component that is independent of the markers (e). If the latter is large then I have problems as I cannot ignore the missing variance. If I have a pedigree I could go back to the idea of polygenic variance to estimate the e with the help of the pedigree. If I have no pedigree I must accept that I will likely have bias as I will not be partitioning all the genetic variance between the markers and will underestimate differences in my original trait.

A second point mentioned above to think about is that if I am using anonymous markers without sequence then capturing the variance of a number of QTLs using LD will require many more markers than QTL. The ideas about the relationship between effects and MAF argued for Van Raden Model (2) becomes rather suspect!

No pedigree, or map or map order is required to fit these models.

Summary

- Can fit multiple markers using shrinkage
  - » Ridge Regression (RR)
- If markers capture total genetic variance
  - » 2 ridge regression models
  - » Model (1) has variance of allelic substitutions independent of MAF
  - » Model (2) with normalised & centred marker design matrix $\mathbf{W}_2$ shares variance among loci
- No pedigree, or map or map order is needed
  - » assume dense across genome to capture σ_A^2

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Summary

- Lead to GBLUP Van Raden Models 1 & 2
  - » Equivalent models to the corresponding RR
  - » GBLUP smaller computing demands than RR

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Summary

- By analogy with **A**, can define
  - » a genomic relationship coefficient
  - » can construct matrix of coefficients, **G**
  - » **G** <u>defined by IBS</u>
  - » there is not a single **G**!
- **G** can replace **A** in models
  - » e.g. animal, repeatability & multi-trait models
    - » QGM lectures 1 to 4
  - » optimum contributions
    - » QG week 10

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Having obtained **G** then can use it in place of **A**, e.g. in repeatability models, multi-trait models etc. One can also use it to replace the **A** in the optimum contribution procedures used to manage diversity as described in QG Week 10.

Additional Optional Material

- 'van_raden_models.pdf'
  - » a more detailed algebraic treatment joining up the SNP BLUP and Van Raden models
- 'meuwissen_hayes_goddard.pdf'
  - » the pioneering paper
- 'van_raden_2008.pdf'
  - » the origin of the relationship matrices

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